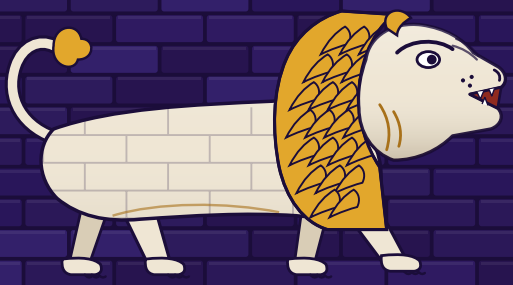
PHYSICS & ENERGY TECHNOLOGY · LEVEL 3

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MECHANICS AND HEAT LAB

A handout of ideas, facts and problems



12 sections · Edition of November 2026 · 39 pages

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How to read this handout

This handout follows the method of Jaan Kalda's physics handouts. The text is short; the work is in the numbered items. Read each section once, then return to its items until you can use them without looking.

Idea 1. An *idea* is a way of thinking that unlocks a whole family of problems. Ideas are framed in violet.

Fact 1. A *fact* is a law or relationship you are expected to know, with the meaning of every symbol and the conditions under which it holds.

Method 1. A *method* is a fixed sequence of steps for a standard task.

Pr 1. Problems are numbered through the handout. A problem followed by a *Solution* is worked for you; cover the solution and try first. For the remaining problems, attempt the problem, then use the *Hints* section, and only then the *Answers* section, both at the back of the subject.

Physical quantities are set in italic, units in roman, with a thin space between number and unit. The index at the end lists every topic and symbol with the page on which it is introduced.

Mechanics and Heat Lab

All distinct experiments in the folder: pendulum, free fall, static and dynamic springs, force table, cooling, Boyle law, expansion, specific heat, latent heat and Joule equivalent. Duplicate versions are consolidated.

1 Earn marks from tables, graphs and uncertainty

Aims. Calculate a fitted-line slope with the correct axis units; Distinguish percentage error from measurement uncertainty.
Before this section: Coordinate differences; division; raw and derived measurements.

A lab result is a measured quantity with units and a justified method. Record raw observations before calculating derived columns. A slope is change in the vertical variable divided by change in the horizontal variable. Its units are essential and determine how to recover the desired constant.

Idea 1. An experimental result combines a number, units and a defensible measurement method. Repeated readings reveal spread, but they may all share the same calibration offset. Percentage error compares a result with a reference; it does not by itself explain whether the discrepancy comes from random variation or systematic bias.

Fact 1.

$$m = \Delta y / \Delta x$$

$$\%\ \text{error} = 100|x_{\text{measured}} - x_{\text{reference}}|/|x_{\text{reference}}|$$

where m is the slope of the selected vertical variable against horizontal variable (vertical unit / horizontal unit); x_{measured} is the experimental estimate (same as reference); $x_{\text{reference}}$ is the accepted comparison value (same as measurement).

Why it holds. Repeating a measurement can reduce random scatter in the mean. A shared calibration offset remains in repeated readings unless it is identified and corrected.

Method 1.

1. Record raw readings with units before calculating derived quantities.
2. Label axes with quantities and units and draw a suitable best-fit line.
3. Choose two widely separated points on that fitted line and calculate rise over run.
4. Use axis units to interpret the slope; compare with a reference using the stated percentage-error formula.

Recognising it. Use percentage error for a reference comparison, and discuss instrument or method bias separately from random spread.

Pr 1. A fitted line rises by 0.8 square seconds over 0.2 m.

Solution. The example has vertical change 0.8 s^2 and horizontal change 0.2 m. Slope is rise over run: $0.8/0.2 = 4$ s^2/m. The reciprocal, 0.25 m/s^2, belongs to a graph with reversed axes and is a different slope. A slope should come from the fitted line, not necessarily from two adjacent raw observations.

Common error. Do not force a line through the origin unless the model and measurement offsets justify it.

In the examination. Write slope units next to the numerical calculation; they help reveal an inverted axis choice.

Pr 2. Can percentage error alone describe repeatability of a measurement?

Pr 3. A measured g is 9.6 m/s^2 and reference g is 9.8 m/s^2. Calculate percentage error and name one possible systematic error.

Pr 4. A ruler has a fixed zero offset. Taking many readings and averaging them will:

(a) Retain the systematic offset unless corrected

(b) Automatically remove the offset

(c) Prove the ruler is accurate if the readings agree

Fact 2. Uncertainty is different from percentage error. An uncertainty describes measurement quality. Percentage error compares a result with a reference value.

$$\text{reported result} = \text{value} \pm \text{uncertainty}$$

$$\%\ \text{error} = 100\frac{|x_{\text{measured}} - x_{\text{reference}}|}{|x_{\text{reference}}|}$$

1. Use the uncertainty method specified by your experiment or instrument; do not invent a precision from extra calculator digits.
2. Give the uncertainty the same unit as the value. Round the value to the same decimal place as the reported uncertainty.
3. Repeated measurements help estimate and reduce the effect of random scatter. They do not remove a consistent zero offset.
4. An uncertainty range is not a guarantee that every possible error is inside it.

Pr 5. A result is (9.6 ± 0.2) m/s^2, and the reference is 9.8 m/s^2. State the given uncertainty and calculate percentage error.

2 Simple pendulum

Aims. Measure pendulum length and period consistently; Recover g from one trial or a T^2-versus-L graph.
Before this section: Complete oscillation; squaring; slope interpretation.

For small angles a pendulum period depends mainly on length and gravity, not bob mass. Measure length from pivot to the centre of the bob. Timing several complete oscillations reduces the relative impact of reaction time. Keep the starting angle small and avoid pushing the bob.

Idea 2. A pendulum period is the time for a complete return, not merely a trip to the other side. Timing many cycles spreads the start-stop reaction uncertainty over a longer total duration. Its length is a geometric quantity from pivot to bob centre, and a consistent length definition is as important as careful timing.

Fact 3.

$$T = t/N$$
$$T^2 = (4\pi^2/g)L$$
$$g = 4\pi^2/m \text{ for } T^2 \text{ versus } L$$

where L is the pivot-to-bob-centre length (m); t is the time for N complete oscillations (s); T is the period of one oscillation (s); g is the gravitational acceleration (m/s^2).
Conditions. The simple model assumes small angular displacement, negligible damping and a light inextensible suspension.

Why it holds. Dividing a multi-cycle time by cycle count gives the mean single-cycle period. For small angles, period depends on length and gravity rather than bob mass.

Method 2.

1. Measure length from pivot to the bob centre and release from a small angle without a push.
2. Time several complete oscillations and divide total time by their count.
3. Calculate T^2 and repeat at several lengths if graphing.
4. For T^2 on the vertical axis and L horizontally, use $g = 4\pi^2/\text{slope}$.

Recognising it. A pendulum time for several cycles first requires conversion to period before applying the gravity relation.

Pr 6. For L=1 m and T=2 s, $g = 4\pi^2 \times 1/4 = 9.87$ m/s^2.

Solution. For the example L = 1 m and T = 2 s. Square the measured period: $T^2 = 4$ s^2. Rearranging $T^2 = 4\pi^2 L/g$ gives $g = 4\pi^2 L/T^2$. Substitution gives $g = 4\pi^2 \times 1/4 \approx 9.87$ m/s^2.

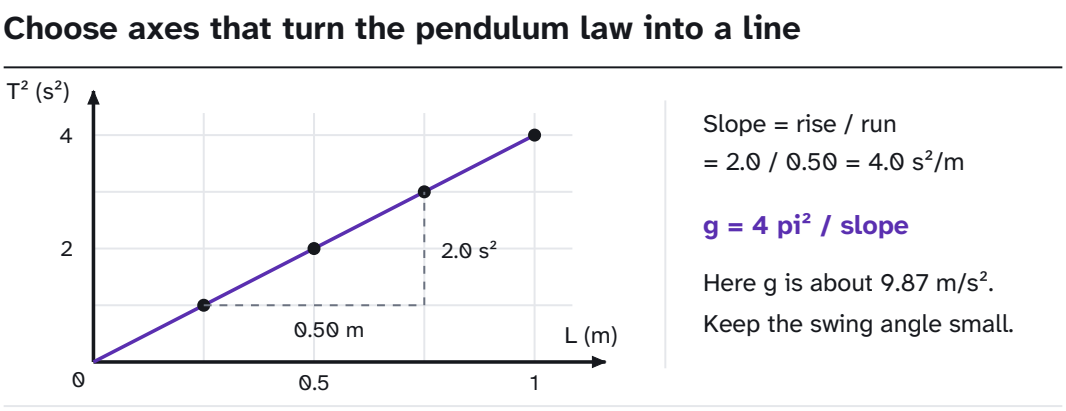


Figure 1. Figure 05. For small pendulum oscillations, a plot of T^2 against L has slope $4\pi^2/g$. The illustrated slope is 4.0 s^2/m, giving g about 9.87 m/s^2. Points are ideal model values, not experimental data.

Common error. Using total timing t instead of the single-cycle period T makes g dramatically wrong.

In the examination. State which variables are on which axes before converting a slope into g.

Pr 7. A bob crosses equilibrium twice in one full cycle. Should each crossing be counted as a cycle?

Pr 8. A pendulum takes 30 s for 15 cycles at length 0.99 m. Calculate period and g.

Pr 9. The pendulum bob is replaced by a heavier bob at the same effective length and small angle. The ideal period:

(a) Halves whenever mass doubles

(b) Stays approximately the same

(c) Increases in proportion to mass

3 Free fall and axis choice

Aims. Linearize free fall with a time-squared graph; Select the correct g conversion for either axis orientation.
Before this section: Constant acceleration from rest; square of time; fitted-line slope.

For release from rest with negligible air drag, fall distance grows with time squared. The course sheets mention both graph orientations; the equation must decide the slope conversion. A light sensor or electronic timer avoids large manual timing errors.

Idea 3. Free fall from rest is quadratic in time, so plotting distance directly against time gives a curve. Squaring time creates a straight-line relationship. Which variable is vertical then determines whether slope represents half of gravity or the reciprocal form; the graph label is part of the calculation.

Fact 4.

$$h = \tfrac{1}{2}gt^2;$$
$$m_{h \text{ vs } t^2} = g/2;$$
$$m_{t^2 \text{ vs } h} = 2/g$$

where h is the vertical fall distance (m); t is the fall time (s); g is the gravitational acceleration (m/s^2); m is the graph slope (depends on axes).
Conditions. The object starts from rest and air resistance is negligible over the measured fall.

Why it holds. For height plotted vertically against time squared, slope has acceleration units. Reversing the axes reverses the ideal slope and changes the conversion needed for gravity.

Method 3.

1. Release from rest and measure fall time with the stated apparatus.
2. Calculate t^2 for each height.
3. State the graph axes and compare h = (g/2)t^2 with y = mx.
4. For h against t^2 use g = 2m; for t^2 against h use g = 2/m.

Recognising it. When a free-fall graph is given, read the axes before recalling any slope-to-gravity rule.

Pr 10. A plot of h against t^2 has slope 4.9 m per square second, so g=9.8 m/s^2.

Solution. With h on the vertical axis, the fitted slope is 4.9 m/s^2. The equation h = (g/2)t^2 shows this slope is half g, so g = 9.8 m/s^2. Reversing the axes makes slope 2/g, about 0.204 s^2/m. The reciprocal slopes describe the same ideal relationship; neither slope can be interpreted without its axes.

Common error. Do not use g=2 times slope for a t-squared-versus-height graph.

In the examination. Do not memorize slope = g/2 without the graph orientation; use the displayed equation to decide.

Pr 11. A t^2-versus-h slope is 0.20 s^2/m. What is g?

Pr 12. The slope of t^2 against height is 0.20 square seconds per metre. Find g.

Pr 13. For release from rest without drag, why is height plotted against time squared?

(a) It removes gravity from the experiment

(b) It makes height proportional to time

(c) It makes the ideal relationship linear

4 Static Hooke law

Aims. Distinguish spring extension from loaded length; Find spring stiffness from a load or a force-extension slope.
Before this section: Weight mg; length differences; centimetres to metres.

Within the proportional range, spring extension is proportional to the applied load. Extension is loaded length minus unloaded length. The restoring spring force points opposite the extension, while the applied load magnitude is k times extension. An elastic spring returns to its original length after unloading, but that alone does not prove a straight-line force-extension relation.

Idea 4. A spring experiment measures extension relative to the unloaded state. The hanging mass supplies a weight, which is a force rather than a length. Within the proportional range, plotting load against extension reveals stiffness: a steep line means more force is needed for each extra metre of stretch.

Fact 5.

$$F = k\Delta L$$
$$F = mg$$
$$k = m_{F \text{ vs } \Delta L}$$

where F is the applied load magnitude (N); ΔL is the loaded minus unloaded spring length (m); k is the spring stiffness (N/m); m is the total suspended mass (kg).
Conditions. Use Hooke law only over the proportional linear range; elastic recovery alone does not guarantee strict proportionality.

Why it holds. Using total loaded length would include an irrelevant original-length offset. Weight must be calculated from mass before dividing by extension to obtain stiffness.

Method 4.

1. Record the unloaded spring length first.
2. Add the load safely and calculate extension as loaded minus unloaded length.

3. Convert extension to metres and load mass to force using mg.
4. Find k from F/ΔL or the slope of F against ΔL; check the graph remains linear.

Recognising it. For static spring problems, convert mass to force and loaded-minus-unloaded length to extension.

Pr 14. A 2 N load produces 0.04 m extension, so k=50 N/m.

Solution. The applied load is 2 N and the extension is 0.04 m, not the total spring length. Using $F = k\Delta L$ gives $k = 2/0.04 = 50$ N/m. This means an additional 1 m extension would require 50 N only if the same linear model remained valid. If unloading leaves a changed length, the spring has not returned elastically; avoid extrapolating the original model.

Common error. Centimetres must be converted to metres before reporting k in N/m.

In the examination. Include the hanger in total suspended mass if the apparatus instructions require it; otherwise the force scale can acquire an offset.

Pr 15. A 0.30 kg total load extends the spring 6 cm with $g = 10$ m/s^2. What is k?

Pr 16. A 0.30 kg mass extends a spring by 6 cm. Use g=10 m/s^2. Find k.

Pr 17. A spring original length is 10 cm and its loaded length is 14 cm. Which length belongs in the Hooke-law extension term?

(a) The 4 cm difference

(b) The 14 cm loaded length

(c) The 10 cm original length

5 Dynamic spring and gravity

Aims. Relate mass-spring period to load and spring stiffness; Use graph axes to distinguish a k experiment from a g experiment.
Before this section: Static spring extension; period measurement; slope units.

An oscillating mass-spring system has a period that increases with mass and decreases with stiffness. Combining static extension with dynamic period allows gravity to be estimated. Small oscillations and the same effective load model are needed for consistency.

Idea 5. Static stretch and dynamic oscillation probe the same spring in different ways. A larger oscillating mass produces a longer period, while a stiffer spring produces a shorter one. Squaring period makes its relationship with mass linear, so the graph slope contains the stiffness information.

Fact 6.

$$T^2 = 4\pi^2 m/k$$
$$T^2 = (4\pi^2/g)\Delta L$$

where T is the oscillation period (s); m is the oscillating mass in the stated model (kg); k is the spring stiffness (N/m); ΔL is the static equilibrium extension (m); g is the gravitational acceleration (m/s^2).
Conditions. Small oscillations stay in the spring proportional range; the static and dynamic models must treat hanger and effective spring mass consistently.

Why it holds. For period squared against mass, a larger slope corresponds to a smaller spring stiffness. A period-squared-versus-extension graph instead contains gravity, so the two slope interpretations cannot be exchanged.

Method 5.

1. Measure small vertical oscillations about equilibrium and obtain period from several cycles.
2. For T^2 against mass compare the slope with $4\pi^2$/k.
3. For T^2 against static extension compare the slope with $4\pi^2$/g.
4. Apply the matching rearrangement and retain the effective-mass assumptions used by the apparatus.

Recognising it. Inspect whether the horizontal axis is mass or static extension before converting a spring-oscillation slope.

Pr 18. A T-squared-versus-extension slope of 4.0 square seconds per metre gives g= $4\pi^2/4.0$=9.87 m/s^2.

Solution. The example plots T^2 vertically against static extension horizontally. Its slope is 4.0 s^2/m, corresponding to $4\pi^2$/g. Rearrangement gives g = $4\pi^2/4.0 \approx 9.87$ m/s^2. A T^2-versus-mass slope would instead determine k; substituting it here would mix two different experiments.

Common error. The symbols for mass and graph slope can both be m; label them explicitly to avoid confusion.

In the examination. Use slope units to identify the requested constant: s^2/kg for stiffness, s^2/m for gravity in these two plots.

Pr 19. A T^2-versus-mass slope is 0.80 s^2/kg. What is k?

Pr 20. A T-squared-versus-mass graph has slope 0.80 square seconds per kg. Find spring constant.

Pr 21. Two springs give period-squared-versus-mass lines. The stiffer spring has:

(a) The same slope regardless of stiffness

(b) The smaller slope

(c) The larger slope

6 Force table and equilibrium

Aims. Calculate the equilibrant from perpendicular forces; Explain how the force-table ring indicates approximate equilibrium.
Before this section: Vector components; Pythagoras; angles from positive x.

The force table demonstrates vector addition. The resultant of two forces is their vector sum. The equilibrant has equal magnitude and opposite direction. A centred ring that does not press on the pin indicates approximate balance, subject to pulley friction.

Idea 6. The force table balances vector directions, not just force magnitudes. Two perpendicular pulls create a resultant pointing between them; a third force balances that resultant only if it points exactly opposite. A centred ring provides experimental evidence of approximate balance when it is not held by the pin.

Fact 7.

$$\sum F_x = 0$$
$$\sum F_y = 0$$
$$F_i = m_i g$$

where F_i is the tension force from hanging mass in the ideal setup (N); m_i is the total hanging mass for a string (kg); R is the resultant force magnitude (N).
Conditions. Mass-to-tension conversion neglects pulley friction and string mass; actual balance is approximate.

Why it holds. Equal resultant and equilibrant magnitudes are not enough without opposite directions. A force pushing the ring against the central pin can be balanced partly by the pin, hiding string-force imbalance.

Method 6.

1. Convert each hanging mass to a force using the chosen g.
2. Resolve all forces into signed horizontal and vertical components.
3. Find the resultant, then reverse both components for the equilibrant.
4. Adjust the balancing force and check the ring is centred without pressing on the pin.

Recognising it. When equilibrium is requested, first find the resultant, then reverse its components for the equilibrant.

Pr 22. For 3 N east and 4 N north, the resultant is 5 N northeast.

Solution. The 3 N east and 4 N north forces give resultant components (+3,+4) N. Its magnitude is $\sqrt{(3^2+4^2)} = 5$ N. Its precise direction is 53.1° north of east; northeast here describes its quadrant. The equilibrant has components (−3,−4) N and magnitude 5 N, directed 53.1° south of west.

Common error. The equilibrant is not the resultant in the same direction.

In the examination. If an angle is requested, northeast alone is insufficient; give a numerical angle and reference axis.

Pr 23. Does the equilibrant have the same direction as the resultant?

Pr 24. Two forces are 6 N east and 8 N north. Find equilibrant magnitude and direction.

Pr 25. Why should the force-table ring be free of contact with the central pin?

(a) The pin removes all pulley friction

(b) A ring touching a pin has no weight

(c) The pin could supply an extra balancing force

7 Newton cooling law

Aims. Calculate temperature excess above the surroundings; Interpret an exponential cooling trend without confusing it with Celsius temperature.
Before this section: Subtracting temperatures; halving; logarithm recognition.

A hot object approaches room temperature as its temperature excess decays. The simple model assumes approximately constant surroundings and an effective cooling constant over the measured range. The temperature excess, not the absolute temperature, is the decaying quantity.

Idea 7. Cooling depends on how much hotter the object is than its surroundings. An object at 50°C in a 20°C room has an excess of 30°C, not 50°C. The simple exponential model shrinks this excess, then room temperature must be added back to recover the actual object temperature.

Fact 8.

$$T - T_r = (T_0 - T_r)e^{-kt}$$
$$\ln[(T - T_r)/(T_0 - T_r)] = -kt$$

where T is the object temperature (°C or K consistently); T_r is the room temperature (same unit as T); k is the effective cooling constant (inverse time); t is the elapsed time (s or min consistently with k).
Conditions. Room temperature is approximately constant and the effective cooling constant applies over the measured range.

Why it holds. Equal temperature excesses in otherwise equivalent conditions give equal cooling-rate magnitudes in the model. As excess falls, the cooling rate becomes smaller and the object approaches room temperature.

Method 7.

1. Measure room temperature and record it as the reference.
2. Subtract room temperature from each object temperature.

3. Compare excesses over time or use the stated logarithmic plot.
4. Add the room temperature back when converting a final excess into an object temperature.

Recognising it. A statement that temperature excess halves means halve object-minus-room temperature, not the Celsius reading itself.

Pr 26. If room temperature is 20 degrees C and the object cools from 80 to 50, the excess falls from 60 to 30 degrees, so it halves.

Solution. The room is at 20°C, so the initial 80°C object has excess $80 - 20 = 60$°C. At 50°C its excess is $50 - 20 = 30$°C. The excess has halved, even though 50 is not half of 80. This distinction explains why the model approaches room temperature rather than cooling toward 0°C.

Common error. The object approaches room temperature, not zero degrees C.

In the examination. When logging data, use a dimensionless excess ratio such as (T-Tr)/(T0-Tr), with temperatures consistently measured.

Pr 27. At room temperature 25°C, an object starts at 85°C and its excess halves. What is its new temperature?

Pr 28. Room temperature is 25 degrees C. A body starts at 85 degrees C and its temperature excess halves in 5 minutes. What is its temperature then?

Pr 29. A 60°C object is placed in a 20°C room or a 40°C room under otherwise identical conditions. It initially cools faster in:

(a) The 20°C room

(b) The 40°C room

(c) Both rooms equally because the object starts at 60°C

8 Boyle law

Aims. Use absolute pressure in a fixed-temperature gas comparison; Predict the inverse pressure-volume trend and check thermal settling.

Before this section: Inverse proportion; multiplying and dividing; consistent pressure and volume units.

At fixed temperature and fixed amount of gas, absolute pressure is inversely proportional to volume. A narrow uniform tube may use gas-column length as a proxy for volume. Let the gas settle thermally after changing pressure.

Idea 8. Boyle law follows the pressure-volume trade-off for a fixed amount of gas at constant temperature. The gas can occupy less space only while exerting greater absolute pressure. Rapid compression can warm the gas, so thermal settling is needed before treating readings as one constant-temperature comparison.

Fact 9.

$$P_1V_1 = P_2V_2$$
$$P \propto 1/V$$

where P is the absolute gas pressure (Pa or consistent pressure unit); V is the gas volume (m^3 or consistent volume unit).
Conditions. Gas amount and temperature remain constant; the ideal Boyle-law approximation is suitable for the measured range.

Why it holds. The pressure-volume product stays constant only while amount and temperature stay fixed. Using gauge pressure instead of absolute pressure changes the comparison unless converted appropriately.

Method 8.

1. Confirm the gas amount stays fixed and readings are taken at the same temperature.
2. Use absolute pressure rather than gauge pressure.
3. Write P1V1 = P2V2 and isolate the requested volume or pressure.
4. Check that larger pressure gives smaller volume and allow time for thermal settling between trials.

Recognising it. Use Boyle law when gas amount is fixed and two states are explicitly at the same temperature.

Pr 30. A gas at 100 kPa and 60 cubic centimetres has volume 30 cubic centimetres at 200 kPa, if temperature stays fixed.

Solution. Initially the gas has pressure 100 kPa and volume 60 cm^3. At fixed temperature and amount, their product stays constant. Doubling pressure to 200 kPa requires half the volume: V2 = 100 × 60/200 = 30 cm^3. Keeping kPa and cm^3 is valid for this ratio because the corresponding units match on both sides.

Common error. Compressing quickly can warm the gas and violate the constant-temperature assumption.

In the examination. Do not convert a gauge reading to absolute pressure by ignoring atmospheric pressure; use the apparatus definition.

Pr 31. If volume is compressed quickly, why wait before recording a Boyle-law reading?

Pr 32. A trapped gas has volume 40 cubic centimetres at absolute pressure 120 kPa. Find its volume at 160 kPa at the same temperature.

Pr 33. A gas is compressed rapidly and its temperature rises. Can the original constant-temperature Boyle comparison be applied immediately?

(a) Only if gauge pressure replaces absolute pressure

(b) No; the temperature condition is not yet satisfied

(c) Yes; any gas compression follows the same constant product

9 Linear thermal expansion

Aims. Calculate a linear expansion coefficient with compatible length units; Extract the coefficient from a length-change-versus-temperature graph.
Before this section: Small length changes; temperature differences; mm-to-m conversion.

Heating changes the length of a rod slightly. The coefficient of expansion describes fractional length change per degree. A dial gauge often reads millimetres while the rod length is recorded in metres, so consistent units are critical.

Idea 9. An expansion coefficient describes fractional length change per degree, so it should not depend on choosing millimetres or metres for the apparatus readout. Problems arise when the tiny displacement is entered in millimetres while the original rod remains in metres. Consistent length units must precede division.

Fact 10.

$$\Delta L = \alpha L_0 \Delta T$$
$$\alpha = m_{\Delta L \text{ vs } \Delta T} / L_0$$

where α is the linear expansion coefficient (K^{-1}); L_0 is the initial rod length (m); ΔL is the length increase (m); ΔT is the temperature increase (K).
Conditions. The temperature range is small enough to treat α as approximately constant and the rod temperature as sufficiently uniform.

Why it holds. The same displacement has different numerical values in metres and millimetres. Dividing length change by original length removes the length unit, leaving inverse temperature.

Method 9.

1. Record original rod length and the initial and final temperatures.
2. Convert gauge displacement to the same length unit as the rod.
3. Rearrange $\Delta L = \alpha L0 \Delta T$ to solve for α.
4. For a graph of ΔL against ΔT divide its slope by L0.

Recognising it. When a dial displacement and rod length use different units, convert them before calculating the coefficient.

Pr 34. A 1 m rod expands 0.5 mm over 50 K, giving alpha = $0.0005/(1 \times 50)$=0.00001 per K.

Solution. The 1 m rod expands by 0.5 mm, which is 0.0005 m. The temperature rise is 50 K. Divide extension by original length and temperature rise: $\alpha = 0.0005/(1 \times 50)$. The result is 0.00001 K^{-1}, or 1.0×10^{-5} K^{-1}.

Common error. A graph against absolute temperature may have an intercept; do not confuse it with a change-of-temperature plot.

In the examination. Write the displacement conversion explicitly; the small numerical size of α is physically expected.

Pr 35. What conversion error occurs if 0.5 mm is entered as 0.5 m?

Pr 36. A 0.50 m rod expands 0.30 mm after warming by 40 K. Find its expansion coefficient.

Pr 37. A student enters a millimetre displacement number as metres while keeping rod length in metres. The coefficient is:

(a) 1000 × too small

(b) Unaffected because coefficients have no length unit

(c) 1000 × too large

10 Specific heat, latent heat and Joule equivalent

Aims. Separate sensible heating, phase-change energy and electrical input; Keep the joule-per-calorie conversion dimensional.
Before this section: Energy balance; electrical power; grams-to-kilograms conversion.

Mixing calorimetry balances heat lost by the hot sample against heat gained by water and calorimeter. Latent heat accounts for a phase change. Electrical heating provides IVt joules; comparing it with heat measured in calories gives joules per calorie, not a dimensionless ratio.

Idea 10. Electrical heating supplies energy over a measured time. If a sample already at its melting point uses all that input only for melting, dividing by melted mass gives latent heat per kilogram. If temperature changes or losses also occur, the same division would misattribute energy to melting.

Fact 11.

$$m_s c_s (T_s - T_f) = (m_w c_w + C_{\text{cal}})(T_f - T_w)$$
$$Q = mL$$
$$E = IVt$$

where Q is the thermal energy transferred (J); L is the specific latent heat (J/kg); I is the heater current (A); V is the heater voltage (V); t is the heating duration (s).
Conditions. The latent-heat example assumes the sample starts at its melting point and all stated energy is used for melting.

Why it holds. Voltage times current gives power, so multiplying by time gives total supplied energy. Latent heat is energy per mass, not total energy and not a temperature rise.

Method 10.

1. Identify whether temperature changes, phase changes or both occur.

2. Calculate electrical input as IVt when heater readings are supplied.
3. Use mcΔT for temperature change and mL for a phase change.
4. If comparing joules with calories, divide joules by calories and state J/cal.

Recognising it. Use the electrical-input and latent-heat relations together only after identifying where the supplied energy goes.

Pr 38. If 500 J melts 0.002 kg at its melting point with no other heating, latent heat is 250000 J/kg.

Solution. The sample receives 500 J only for melting and the melted mass is 0.002 kg. Rearranging $Q = mL$ gives $L = 500/0.002 = 250000$ J/kg. In the separate conversion example, 418.4 J is compared with 100 cal. Their ratio is $418.4/100 = 4.184$ J/cal; the unit is essential because the two energy scales differ.

Common error. Ignoring calorimeter heat capacity or heat loss can bias the inferred material property.

In the examination. State where the energy goes before dividing by mass; not all heating energy necessarily produces the phase change.

Pr 39. If both electrical input and thermal gain are already in joules, is their ratio a J/cal conversion?

Pr 40. A heater at 6 V carries 2 A for 100 s. If all energy melts a sample already at its melting point and the melted mass is 0.004 kg, find supplied energy and latent heat.

Pr 41. Why must the sample already be at its melting point in the simple electrical melting calculation?

(a) Otherwise some input may first warm the sample

(b) Otherwise electrical energy cannot enter matter

(c) Because latent heat means temperature rise per second

11 Calorimetry: determine a solid specific heat

Aims. Balance heat lost by a solid against heat gained by water and calorimeter; Solve for specific heat while using the correct temperature changes.

Before this section: Heat capacity; mass in kg; a common final equilibrium temperature.

A hot solid placed in cooler water transfers energy to the water and calorimeter. At equilibrium they share a final temperature. The ideal model assumes negligible heat loss to surroundings and no phase change. The calorimeter contribution must be included if it is specified.

Idea 11. Calorimetry matches energy leaving a hot sample with energy entering cooler water and, if relevant, its container. Each body has its own temperature change despite reaching one shared final temperature. Writing those two changes explicitly is often the most important step.

Fact 12.

$$c_s = \frac{(m_w c_w + C_{\text{cal}})(T_f - T_w)}{m_s(T_s - T_f)}$$

where c_s is the solid specific heat capacity (J/(kg K)); m_s, m_w is the solid and water masses (kg); C_{cal} is the calorimeter heat capacity (J/K); T_f is the final equilibrium temperature (°C or K).
Conditions. The example neglects surroundings losses and calorimeter heat capacity and assumes no phase change.

Why it holds. The solid temperature drop determines how much energy it loses for each unit of its specific heat. The water temperature rise and known specific heat determine the energy gained in the ideal balance.

Method 11.

1. Identify the initially hot body, initially cold water and common final temperature.
2. Write heat lost by the solid equal to heat gained by water plus calorimeter.
3. Use positive temperature drops and rises on opposite sides.
4. Solve for cs and state the heat-loss and calorimeter assumptions.

Recognising it. Use a heat-lost equals heat-gained balance when a hot solid and cooler water reach one final temperature.

Pr 42. With a negligible calorimeter, 0.10 kg solid cooling from 80 to 30 degrees C heats 0.20 kg water from 25 to 30 degrees C.

Solution. Water warms from 25°C to 30°C, a rise of 5 K. Its heat gain is 0.20 × 4200 × 5 = 4200 J. The 0.10 kg solid cools from 80°C to 30°C, a drop of 50 K. With negligible calorimeter heat capacity and losses, 0.10 × cs × 50 = 4200, giving cs = 840 J/(kg K).

Common error. Using the same temperature change for hot solid and water is generally wrong.

In the examination. Use the calorimeter heat-capacity term when supplied; discarding it undercounts heat gained by the cold side.

Pr 43. Why is the solid temperature change not the same as the water temperature change?

Pr 44. A 0.20 kg solid cools from 90 to 40 degrees C and warms 0.10 kg water from 30 to 40 degrees C. Ignore calorimeter heat capacity and losses; use water c=4200 J/(kg K). Find the solid specific heat.

Pr 45. If a calorimeter warms but its heat gain is ignored, what happens to the inferred solid specific heat under this calculation?

(a) It cannot change because water mass is unchanged

(b) It is underestimated

(c) It is overestimated

12 Latent heat and electrical-to-thermal conversion

Aims. List all energy stages for ice melting and subsequent warming; Interpret how lost heat biases a measured electrical-to-calorie ratio.
Before this section: Specific heat versus latent heat; phase-change temperatures; units of energy.

A phase-change experiment separates melting from subsequent warming of the melted material. In the Joule-equivalent experiment electrical energy is compared with heat gained by water and calorimeter. If both quantities are already in joules, their ratio tests energy accounting; it is not a joule-per-calorie conversion.

Idea 12. A joule-per-calorie experiment compares the same intended energy transfer measured on two unit scales. If some supplied energy escapes without appearing in measured heat gain, the denominator becomes too small. Distinguishing units from efficiency prevents treating every energy ratio as a unit conversion.

Fact 13.

$$Q_{\text{ice}} = mL_f + mc_w(T_f - 0°\text{C})$$
$$J_{\text{eq}} = E_{\text{joules}}/Q_{\text{calories}}$$

where m is the ice mass (kg); L_f is the specific latent heat of fusion (J/kg); c_w is the liquid water specific heat (J/(kg K)); J_{eq} is the joules per calorie conversion estimate (J/cal).
Conditions. The ice formulas use an ordinary-pressure melting point of approximately 0°C and account for the stated final phase.

Why it holds. Joules divided by calories has a conversion unit. Joules divided by joules is dimensionless and describes energy accounting rather than a joule-per-calorie conversion.

Method 12.

1. Identify whether ice starts below or at its melting point.
2. Include warming of ice if needed, then melting, then warming of meltwater to the final temperature.
3. For conversion, keep electrical energy in joules and measured thermal gain in calories.
4. Assess a loss by asking whether the recorded heat gain is too small or too large for the supplied energy.

Recognising it. Check the denominator units before interpreting any electrical-to-thermal energy ratio.

Pr 46. If ice begins below zero degrees C, an additional warming-of-ice term is required before melting.

Solution. Ice below 0°C first needs energy to warm to 0°C; that stage uses ice specific heat. At 0°C, energy mLf melts it without a temperature rise in the ideal fixed-pressure model. If the final state is liquid above 0°C, add mcw(Tf - 0°C) for the meltwater. When ice al-

ready starts at 0°C the first stage is absent; the melting and later warming stages remain if required.

Common error. Do not divide joules by joules and label the dimensionless answer J/cal.

In the examination. Do not use the water specific heat for warming solid ice; the two phases have different heat capacities.

Pr 47. If 836.8 J supplies a recorded heat gain of 200 cal, what is E/Q, and how would unrecorded heat loss bias it?

Pr 48. An electrical experiment supplies 836.8 J and records heat gain of 200 cal. Find the conversion factor. If unaccounted heat escapes, does E/Q measured tend to be too high or too low?

Pr 49. Both electrical input and measured thermal gain are stated in joules. Their ratio has:

(a) Units of J/cal automatically

(b) Units of watts

(c) No unit, so it is not directly a J/cal conversion

Hints

Use one hint at a time, then return to the problem.

Pr 3 (p. 3) (1) Compare the measured and reference values using an absolute difference; the denominator must be the reference. (2) Divide that difference by 9.8 m/s^2 and multiply by 100 percent. (3) For the second part name a specific fixed bias, such as a timing calibration offset, and explain why it would affect readings consistently.

Pr 5 (p. 3) (1) The stated uncertainty is 0.2 m/s^2. (2) Percentage error is $100|9.6-9.8|/9.8 \approx 2.04\%$.

Pr 8 (p. 5) (1) Use the 30 s reading as the total for 15 complete cycles, not the single period. (2) Divide total time by cycle count and square the resulting period. (3) Use length 0.99 m in the rearranged pendulum relation for gravity. Check that the squared period is in seconds squared.

Pr 12 (p. 7) (1) The stated graph puts time squared vertically and height horizontally. Write the fall equation with that vertical quantity isolated. (2) Compare it with a straight-line equation to identify the slope as the reciprocal-gravity form. (3) Use the slope 0.20 s^2/m to solve for gravity. Verify that the resulting units are m/s^2.

Pr 16 (p. 8) (1) The 0.30 kg mass supplies a weight; the 6 cm measurement is its extension. (2) Convert extension to metres and multiply mass by the specified gravity. (3) Divide the applied force by extension to obtain stiffness. Retain N/m and assume the spring remains in its proportional range.

Pr 20 (p. 9) (1) The horizontal axis is mass, so choose the stiffness relation rather than the extension-based gravity relation. (2) Write the slope coefficient of mass in the squared-period equation. (3) Rearrange to make stiffness the subject and substitute 0.80 s^2/kg. Check that the result has N/m units.

Pr 24 (p. 11) (1) Draw 6 N east and 8 N north as perpendicular components. (2) Find their resultant magnitude and its reference angle from the component ratio. (3) Reverse both components to obtain the equilibrant. Give a numerical angle with a compass reference, not merely southwest.

Pr 28 (p. 12) (1) Subtract room temperature 25°C from the initial 85°C reading to obtain excess. (2) Halve that excess for the stated 5-minute interval. (3) Add the unchanged room temperature back. Check that the result lies between initial object temperature and room temperature.

Pr 32 (p. 13) (1) Check that the supplied 120 and 160 kPa are absolute pressures and temperatures match. (2) Set the initial pressure-volume product equal to the final product, then isolate final volume. (3) Use 120 kPa, 40 cm^3 and 160 kPa consistently. The larger final pressure should give a smaller final volume.

Pr 36 (p. 15) (1) The 0.30 mm reading is a length increase, not final rod length. (2) Convert it to metres, then divide by original length 0.50 m and temperature rise 40 K. (3) Report an inverse-kelvin coefficient. Its small magnitude is expected for a typical solid.

Pr 40 (p. 16) (1) Find heater power from 6 V and 2 A before using the 100 s duration. (2) Multiply power by duration to obtain input energy in joules. (3) Because all input melts 0.004 kg at the melting point, divide that energy by melted mass for latent heat. Keep the two answers separately labelled.

Pr 44 (p. 17) (1) Write the solid temperature drop and water temperature rise as separate positive differences. (2) Calculate water heat gain using 0.10 kg and its given specific heat; ignore the calorimeter only because the question says so. (3) Equate that heat to solid mass 0.20 kg times unknown specific heat times its cooling drop, then isolate specific heat.

Pr 48 (p. 19) (1) Use electrical energy as the numerator and thermal gain in calories as the denominator. (2) Divide 836.8 J by 200 cal and retain the unit J/cal. (3) For the loss question, keep input fixed and ask what happens to the ratio if the measured heat-gain denominator becomes too small.

Answers

Pr 2 (p. 3) No. It compares with a reference; repeated-reading spread or an uncertainty estimate is needed to discuss repeatability.

Pr 3 (p. 3) Percentage error = $100 \times 0.2/9.8 = 2.04$ percent. A consistent height-scale zero offset is one possible systematic error. Two marks for the calculation and two for a specific error with its consistency explained.

Pr 4 (p. 3) (a) Correct. Every reading carries the same bias, so averaging alone preserves it.

Pr 5 (p. 3) Uncertainty: $\pm 0.2\,\mathrm{m/s^2}$. Percentage error: about 2.04%. They answer different questions.

Pr 7 (p. 5) No. Count a return to the same position moving in the same direction as one full cycle.

Pr 8 (p. 5) T=30/15=2 s. g=$4\pi^2 \times 0.99/4$=9.77 $\mathrm{m/s^2}$. Two marks per calculation.

Pr 9 (p. 5) (b) Correct. Bob mass cancels in the small-angle simple-pendulum model.

Pr 11 (p. 7) 10 $\mathrm{m/s^2}$, using g = 2/0.20.

Pr 12 (p. 7) g = 2/0.20 = 10 $\mathrm{m/s^2}$. Four marks for axis identification, rearrangement, substitution and unit.

Pr 13 (p. 7) (c) Correct. Distance from rest is proportional to time squared.

Pr 15 (p. 8) 50 N/m: the force is 3 N and extension is 0.06 m.

Pr 16 (p. 8) Force = 3 N; extension = 0.06 m; k=3/0.06=50 N/m. Award marks for force, conversion, ratio and unit.

Pr 17 (p. 8) (a) Correct. Extension is the change from the unloaded length.

Pr 19 (p. 9) About 49.35 N/m, from $4\pi^2/0.80$.

Pr 20 (p. 9) Slope = $4\pi^2/\mathrm{k}$, so k=$4\pi^2$/0.80=49.35 N/m. Credit the selected axes, rearrangement, numerical value and units.

Pr 21 (p. 9) (b) Correct. Stiffness appears inversely in this slope.

Pr 23 (p. 11) No. It has equal magnitude and opposite direction.

Pr 24 (p. 11) Resultant magnitude is 10 N at 53.1 degrees north of east. Equilibrant is 10 N at 233.1 degrees from positive x, or 53.1 degrees south of west. Two marks for magnitude and two for direction.

Pr 25 (p. 11) (c) Correct. Contact can hide an imbalance among the intended string forces.

Pr 27 (p. 12) 55°C: initial excess 60°C becomes 30°C, then add 25°C.

Pr 28 (p. 12) Initial excess = 60 degrees C. Halved excess = 30 degrees C. Add room temperature: 25+30=55 degrees C. Credit both differences and final interpretation.

Pr 29 (p. 12) (a) Correct. Its initial temperature excess is larger in the colder room.

Pr 31 (p. 13) Compression can change temperature; waiting helps the gas return to the intended common temperature.

Pr 32 (p. 13) V2 = $120 \times 40/160 = 30$ cubic centimetres. Credit controlled conditions, equation, rearrangement and unit.

Pr 33 (p. 13) (b) Correct. Wait for the specified thermal condition or use an appropriate different model.

Pr 35 (p. 15) The calculated expansion coefficient becomes 1000 × too large.

Pr 36 (p. 15) Convert expansion to 0.00030 m. Alpha=0.00030/(0.50 × 40)=0.000015 per K. Credit conversion, formula, substitution and result.

Pr 37 (p. 15) (c) Correct. The entered displacement is 1000 times larger than the true metre value.

Pr 39 (p. 16) No. It is dimensionless and tests energy accounting or an efficiency interpretation under the model.

Pr 40 (p. 16) Energy=6 × 2 × 100=1200 J. Latent heat=$\frac{1200}{0.004}$ = 300000 J/kg. Two marks per calculation; the no-loss and initial-phase-temperature assumptions matter.

Pr 41 (p. 16) (a) Correct. Pre-melting warming requires a separate energy term.

Pr 43 (p. 17) They have different starting temperatures, though they reach the same final equilibrium temperature.

Pr 44 (p. 17) Water gains 0.10 × 4200 × 10=4200 J. Solid loses 0.20 × cs × 50=10cs. Thus cs=420 J/(kg K). Two marks for each side and result.

Pr 45 (p. 17) (b) Correct. Ignoring the container undercounts energy lost by the solid, giving too small a specific heat.

Pr 47 (p. 19) 4.184 J/cal. If heat escapes and measured Q is too small, E/Q tends to be too high.

Pr 48 (p. 19) Factor=836.8/200=4.184 J/cal. Heat escaping reduces measured Q for the supplied E, tending to make the ratio too high. Two marks for each part.

Pr 49 (p. 19) (c) Correct. Like energy units cancel.

Mechanics and Heat Lab, Test A

30 marks. Suggested time 30 minutes. Multiple-choice questions carry 1 mark each, written problems 4 marks each. Work without notes.

Part I. Choose one answer.

1. If y is force in N and x is extension in m, slope units are:

   (a) N/m
   (b) m/N
   (c) N × m
   (d) No units

2. Pendulum length should be measured to:

   (a) The top of the table
   (b) The lowest swing point
   (c) The bottom of the bob
   (d) The centre of the bob

3. For t^2 vertically versus h horizontally, g equals:

   (a) Slope squared
   (b) Slope/2
   (c) 2 times slope
   (d) 2/slope

4. The spring constant is the slope of:

   (a) Time versus temperature
   (b) Extension versus force
   (c) Force versus extension
   (d) Mass versus time

5. For fixed k, multiplying the oscillating mass by four makes the period:

   (a) Four times
   (b) Half
   (c) Twice
   (d) Unchanged

6. The equilibrant of a 7 N eastward resultant is:

   (a) 7 N east
   (b) 14 N west
   (c) Zero
   (d) 7 N west

Part II. Show your reasoning.

1. A measured g is 9.6 m/s^2 and reference g is 9.8 m/s^2. Calculate percentage error and name one possible systematic error.

2. A pendulum takes 30 s for 15 cycles at length 0.99 m. Calculate period and g.

3. The slope of t^2 against height is 0.20 square seconds per metre. Find g.

4. A 0.30 kg mass extends a spring by 6 cm. Use g=10 m/s^2. Find k.

5. A T-squared-versus-mass graph has slope 0.80 square seconds per kg. Find spring constant.

6. Two forces are 6 N east and 8 N north. Find equilibrant magnitude and direction.

Mechanics and Heat Lab, Test B

30 marks. Suggested time 30 minutes. Multiple-choice questions carry 1 mark each, written problems 4 marks each. Work without notes.

Part I. Choose one answer.

1. In the simple cooling model, the cooling rate is proportional to:

   (a) Mass only
   (b) Zero
   (c) Temperature excess above surroundings
   (d) Absolute time only

2. Boyle law requires:

   (a) Fixed temperature and gas amount
   (b) Changing gas amount
   (c) Fixed pressure
   (d) Fixed volume

3. Two rods are made of the same material and have the same temperature rise. A rod twice as long has expansion:

   (a) Twice as large
   (b) Unchanged
   (c) Four times as large
   (d) Half as large

4. Electrical energy supplied by constant voltage and current for time t is:

   (a) V/I
   (b) IVt
   (c) I/t
   (d) Vt/I

5. In an ideal mixing experiment, the final equilibrium temperature lies:

   (a) Below both
   (b) Between the initial temperatures
   (c) Always at zero
   (d) Above both initial temperatures

6. To determine joules per calorie, divide electrical energy in joules by heat measured in:

   (a) Watts
   (b) Calories
   (c) Kelvin
   (d) Joules

Part II. Show your reasoning.

1. Room temperature is 25 degrees C. A body starts at 85 degrees C and its temperature excess halves in 5 minutes. What is its temperature then?

2. A trapped gas has volume 40 cubic centimetres at absolute pressure 120 kPa. Find its volume at 160 kPa at the same temperature.

3. A 0.50 m rod expands 0.30 mm after warming by 40 K. Find its expansion coefficient.

4. A heater at 6 V carries 2 A for 100 s. If all energy melts a sample already at its melting point and the melted mass is 0.004 kg, find supplied energy and latent heat.

5. A 0.20 kg solid cools from 90 to 40 degrees C and warms 0.10 kg water from 30 to 40 degrees C. Ignore calorimeter heat capacity and losses; use water c=4200 J/(kg K). Find the solid specific heat.

6. An electrical experiment supplies 836.8 J and records heat gain of 200 cal. Find the conversion factor. If unaccounted heat escapes, does E/Q measured tend to be too high or too low?

Mechanics and Heat Lab, Test A: answers

Suggested marking points are for self-study; your lecturer may use a different scheme.

Question 1. (a) Slope units are vertical units divided by horizontal units.

Written part. Percentage error = $100 \times 0.2/9.8 = 2.04$ percent. A consistent height-scale zero offset is one possible systematic error. Two marks for the calculation and two for a specific error with its consistency explained. (Section: p. 2.)

Question 2. (d) The ideal mass is represented at the bob centre.

Written part. T=30/15=2 s. g=$4\pi^2 \times 0.99/4$=9.77 m/s^2. Two marks per calculation. (Section: p. 4.)

Question 3. (d) Rearranging h = gt squared/2 gives $t^2 = (2/g)$h.

Written part. g = 2/0.20 = 10 m/s^2. Four marks for axis identification, rearrangement, substitution and unit. (Section: p. 6.)

Question 4. (c) Force = k times extension matches y = mx.

Written part. Force = 3 N; extension = 0.06 m; k=3/0.06=50 N/m. Award marks for force, conversion, ratio and unit. (Section: p. 7.)

Question 5. (c) Period is proportional to the square root of mass.

Written part. Slope = $4\pi^2$/k, so k=$4\pi^2$/0.80=49.35 N/m. Credit the selected axes, rearrangement, numerical value and units. (Section: p. 8.)

Question 6. (d) Equal magnitude and opposite direction produce zero net force.

Written part. Resultant magnitude is 10 N at 53.1 degrees north of east. Equilibrant is 10 N at 233.1 degrees from positive x, or 53.1 degrees south of west. Two marks for magnitude and two for direction. (Section: p. 10.)

Mechanics and Heat Lab, Test B: answers

Suggested marking points are for self-study; your lecturer may use a different scheme.

Question 1. (c) Newton cooling uses the difference between object and surrounding temperatures.

Written part. Initial excess = 60 degrees C. Halved excess = 30 degrees C. Add room temperature: 25+30=55 degrees C. Credit both differences and final interpretation. (Section: p. 11.)

Question 2. (a) Only pressure and volume are varied in the ideal Boyle experiment.

Written part. V2 = 120 × 40/160 = 30 cubic centimetres. Credit controlled conditions, equation, rearrangement and unit. (Section: p. 12.)

Question 3. (a) Expansion is proportional to original length.

Written part. Convert expansion to 0.00030 m. Alpha=0.00030/(0.50 × 40)=0.000015 per K. Credit conversion, formula, substitution and result. (Section: p. 14.)

Question 4. (b) Electrical power is VI; multiplying by time gives energy.

Written part. Energy=6 × 2 × 100=1200 J. Latent heat=$\frac{1200}{0.004}$ = 300000 J/kg. Two marks per calculation; the no-loss and initial-phase-temperature assumptions matter. (Section: p. 15.)

Question 5. (b) Energy flows from the hotter body to the cooler one until equilibrium.

Written part. Water gains 0.10 × 4200 × 10=4200 J. Solid loses 0.20 × cs × 50=10cs. Thus cs=420 J/(kg K). Two marks for each side and result. (Section: p. 16.)

Question 6. (b) The denominator determines the per-calorie unit.

Written part. Factor=836.8/200=4.184 J/cal. Heat escaping reduces measured Q for the supplied E, tending to make the ratio too high. Two marks for each part. (Section: p. 18.)

Facts at a glance

1. Earn marks from tables, graphs and uncertainty p. 2

$$m = \Delta y / \Delta x$$
$$\%\ \text{error} = 100|x_{\text{measured}} - x_{\text{reference}}|/|x_{\text{reference}}|$$

2. Simple pendulum p. 4

$$T = t/N$$
$$T^2 = (4\pi^2/g)L$$
$$g = 4\pi^2/m \text{ for } T^2 \text{ versus } L$$

3. Free fall and axis choice p. 6

$$h = \tfrac{1}{2}gt^2;$$
$$m_{h \text{ vs } t^2} = g/2;$$
$$m_{t^2 \text{ vs } h} = 2/g$$

4. Static Hooke law p. 7

$$F = k\Delta L$$
$$F = mg$$
$$k = m_{F \text{ vs } \Delta L}$$

5. Dynamic spring and gravity p. 8

$$T^2 = 4\pi^2 m/k$$
$$T^2 = (4\pi^2/g)\Delta L$$

6. Force table and equilibrium p. 10

$$\sum F_x = 0$$
$$\sum F_y = 0$$
$$F_i = m_i g$$

7. Newton cooling law p. 11

$$T - T_r = (T_0 - T_r)e^{-kt}$$
$$\ln[(T - T_r)/(T_0 - T_r)] = -kt$$

8. Boyle law p. 12

$$P_1 V_1 = P_2 V_2$$
$$P \propto 1/V$$

9. Linear thermal expansion p. 14

$$\Delta L = \alpha L_0 \Delta T$$
$$\alpha = m_{\Delta L \text{ vs } \Delta T} / L_0$$

10. Specific heat, latent heat and Joule equivalent p. 15

$$m_s c_s (T_s - T_f) = (m_w c_w + C_{\text{cal}})(T_f - T_w)$$
$$Q = mL$$
$$E = IVt$$

11. Calorimetry: determine a solid specific heat p. 16

$$c_s = \frac{(m_w c_w + C_{\text{cal}})(T_f - T_w)}{m_s (T_s - T_f)}$$

12. Latent heat and electrical-to-thermal conversion p. 18

$$Q_{\text{ice}} = mL_f + mc_w(T_f - 0°\text{C})$$
$$J_{\text{eq}} = E_{\text{joules}} / Q_{\text{calories}}$$

Notation

Symbols are listed by section. The same letter may name different quantities in different sections; read the section's Fact.

1. Earn marks from tables, graphs and uncertainty

m	slope of the selected vertical variable against horizontal variable (vertical unit / horizontal unit)
x_{measured}	experimental estimate (same as reference)
$x_{\text{reference}}$	accepted comparison value (same as measurement)

2. Simple pendulum

L	pivot-to-bob-centre length (m)
t	time for N complete oscillations (s)
T	period of one oscillation (s)
g	gravitational acceleration (m/s^2)

3. Free fall and axis choice

h	vertical fall distance (m)
t	fall time (s)
g	gravitational acceleration (m/s^2)
m	graph slope (depends on axes)

4. Static Hooke law

F	applied load magnitude (N)
ΔL	loaded minus unloaded spring length (m)
k	spring stiffness (N/m)
m	total suspended mass (kg)

5. Dynamic spring and gravity

T	oscillation period (s)
m	oscillating mass in the stated model (kg)
k	spring stiffness (N/m)
ΔL	static equilibrium extension (m)
g	gravitational acceleration (m/s^2)

6. Force table and equilibrium

F_i	tension force from hanging mass in the ideal setup (N)
m_i	total hanging mass for a string (kg)
R	resultant force magnitude (N)

7. Newton cooling law

T	object temperature (°C or K consistently)
T_r	room temperature (same unit as T)
k	effective cooling constant (inverse time)
t	elapsed time (s or min consistently with k)

8. Boyle law

P	absolute gas pressure (Pa or consistent pressure unit)
V	gas volume (m^3 or consistent volume unit)

9. Linear thermal expansion

α	linear expansion coefficient (K^{-1})
L_0	initial rod length (m)
ΔL	length increase (m)
ΔT	temperature increase (K)

10. Specific heat, latent heat and Joule equivalent

Q	thermal energy transferred (J)
L	specific latent heat (J/kg)
I	heater current (A)
V	heater voltage (V)
t	heating duration (s)

11. Calorimetry: determine a solid specific heat

c_s	solid specific heat capacity (J/(kg K))
m_s, m_w	solid and water masses (kg)
C_{cal}	calorimeter heat capacity (J/K)
T_f	final equilibrium temperature (°C or K)

12. Latent heat and electrical-to-thermal conversion

m	ice mass (kg)
L_f	specific latent heat of fusion (J/kg)
c_w	liquid water specific heat (J/(kg K))
J_{eq}	joules per calorie conversion estimate (J/cal)

Sources

All distinct experiments in the folder: pendulum, free fall, static and dynamic springs, force table, cooling, Boyle law, expansion, specific heat, latent heat and Joule equivalent. Duplicate versions are consolidated.

The source audit inventories 159 non-programming files; 95 yielded substantial extracted text. Scanned diagrams and handwritten or image-only material were not fully transcribed. Check lecturer-specific derivations and conventions. The practice tests are original study material, not predictions of an examination.

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